

5.5 Test Cases

User Login Module

1. Functional Test cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-UL-01	Valid Login	Enter correct email & password → Click Login	Redirect to User Dashboard	Pass
TC-UL-02	Invalid Password	Enter correct email & wrong password	Error message: "Invalid Credentials"	Pass
TC-UL-03	Unregistered Email	Enter email not in database	Error message displayed	Pass
TC-UL-04	Empty Fields	Leave email & password blank	Validation message shown	Pass
TC-UL-05	Partial Input	Enter only email	Password required message	Pass

2. Security Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-UL-06	SQL Injection	Enter ' OR 1=1 -- in login field	Login fails	Pass
TC-UL-07	XSS Attack	Enter <script>alert()</script>	Script not executed	Pass
TC-UL-08	Direct Dashboard Access	Access /dashboard without login	Redirect to login page	Pass
TC-UL-09	Password Storage Check	Check database password field	Password stored in hashed format	Pass
TC-UL-10	Session Hijacking	Modify session manually	Access denied	Pass

3.Session Management test case

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-UL-11	Session Timeout	Login → Stay idle for session time	Auto logout	Pass
TC-UL-12	Logout Function	Click Logout	Session destroyed & redirect to login	Pass
TC-UL-13	Back Button After Logout	Logout → Press back button	Cannot access dashboard	Pass

4.Performance Test cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Status	
TC-UL-14	Login Response Time	Enter valid credentials	Login within 2 seconds	Pass	
TC-UL-15	Concurrent Login	20 users log in simultaneously	System remains stable	Pass	

5.Edge case test cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-UL-16	Very Long Input	Enter 500 characters in email field	Validation error	Pass
TC-UL-17	Special Characters	Enter special characters	Validation error	Pass
TC-UL-18	Case Sensitivity	Enter uppercase email	Login successful (if system ignores case)	Pass

Authentication Module Test Cases

Admin login

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-19	Login with empty fields	Leave username & password blank → Click Login	Validation message displayed	Pass
TC-20	SQL Injection attempt	Enter ' OR 1=1 -- in username	Login fails, system remains secure	Pass
TC-21	Session Expiry	Login → Wait for session timeout → Access dashboard	Redirect to login page	Pass
TC-22	Direct URL Access	Access /admin/dashboard without login	Access denied	Pass

Product Management Module Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-23	Large Image Upload	Upload image greater than allowed size	Upload rejected	Pass
TC-24	Duplicate Product Entry	Add same product twice	System prevents duplication	Pass
TC-25	Invalid Price Format	Enter alphabets in price field	Validation error shown	Pass
TC-26	Missing Required Field	Leave product details empty	Error message displayed	Pass

Product Management Module Test Cases

Update Product

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-27	Update Product Details	Change price → Save	Updated in database	Pass
TC-28	Update Without Permission	Unauthorized user tries update	Access denied	Pass
TC-29	Update with Invalid Data	Enter negative price	Validation error	Pass

Delete Product

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-30	Delete Existing Product	Click Delete	Product removed from database	Pass
TC-31	Delete Non-existing ID	Modify ID in URL	Error handled properly	Pass
TC-32	Delete Product Image	Delete product	Image removed from server	Pass

Image Upload Module Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-33	Upload JPG Image	Upload valid JPG file	Successfully stored	Pass
TC-34	Upload PNG Image	Upload valid PNG file	Successfully stored	Pass
TC-35	Upload Invalid File	Upload .exe file	Upload rejected	Pass
TC-36	Multiple Images Upload	Upload multiple images	All stored correctly	Pass

Contact Module Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-37	Valid Inquiry Submission	Fill form correctly	Data saved in database	Pass
TC-38	Invalid Email Format	Enter incorrect email	Error message shown	Pass
TC-39	Long Message Input	Enter 1000+ characters	Accepted within limit	Pass
TC-40	Spam Protection	Submit form repeatedly	Requests limited or blocked	Pass

Search & Filter Functionality

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-41	Search by Category	Enter "Shirts"	Relevant products shown	Pass
TC-42	Filter by Price Range	Select ₹500–₹1500	Filtered results shown	Pass
TC-43	Search Invalid Keyword	Enter random text	No results found message	Pass
TC-44	Combined Filter	Category + Price + Size	Accurate filtered results	Pass

Dashboard & Analytics

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-45	Dashboard Load	Login as Admin	Stats displayed correctly	Pass
TC-46	Product Count Accuracy	Compare DB count with dashboard	Numbers match	Pass
TC-47	Chart Data Accuracy	Verify monthly data chart	Chart reflects DB data	Pass

Performance Testing (Database & Backend)

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-48	Database Connection Failure	Stop MySQL → Start server	Error handled gracefully	Pass
TC-49	API Response Time	Send API request	Response < 2 seconds	Pass
TC-50	Concurrent Requests	Multiple users access website	No crash	Pass

Security Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-51	XSS Attack	Enter <script>alert()</script> in form	Script not executed	Pass
TC-52	CSRF Protection	Try cross-site request	Request rejected	Pass
TC-53	Password Hash Check	Check database password field	Stored in hashed format	Pass
TC-54	Environment Variable Security	Try accessing DB credentials publicly	Not visible	Pass

Load & Performance Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-55	Load Test (50 users)	Simulate multiple users	System remains stable	Pass
TC-56	Page Load Speed	Open homepage	Loads within 3 seconds	Pass
TC-57	Image Optimization	Load product gallery page	Images optimized	Pass

Responsive Design Testing

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-58	Mobile View	Open on mobile device	Layout is responsive	Pass
TC-59	Tablet View	Open on tablet	UI adjusts correctly	Pass
TC-60	Cross-Browser	Test in Chrome/Edge	Works correctly	Pass

Hosting & Deployment Testing

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-61	Production Environment	Deploy on hosting server	Server runs successfully	Pass
TC-62	SSL Verification	Open HTTPS website	Secure connection established	Pass
TC-63	NPM Install	Run npm install on server	Dependencies installed	Pass